INCH-POUND

MIL-DTL-6669D
10 May 2011
SUPERSEDING
MIL-I-6669C
22 December 1972

MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE, MULTI-FUNCTION, -70 °C TO +150 °C

Inactive for new design after 24 March 1999.

This specification is approved for use by all departments and agencies of the Department of Defense.

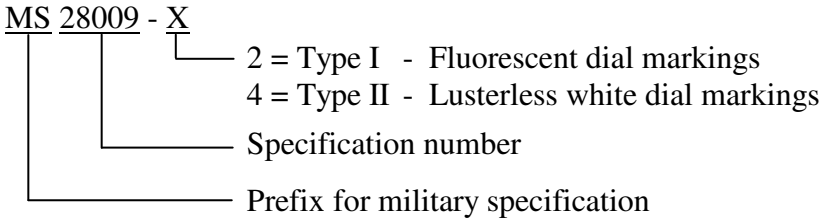
1. SCOPE

1.1 Scope. This specification covers multi-function temperature indicators with a range from -70° to +150° Centigrade (C).

1.2 Classification. The indicators are of 2 classes (see 6.2).

Type I - Fluorescent dial markings (MS28009-2)
Type II - Lusterless white dial markings (MS28009-4)

1.3 Part or identifying number (PIN). The PIN to be used for multi-function temperature indicators acquired to this specification is shown in MS28009.



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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

FEDERAL STANDARDS

FED-STD-595/37875 - Miscellaneous, Flat or Lusterless
FED-STD-595/37038 - Miscellaneous, Flat or Lusterless

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-I-7070	- Indicator, Temperature, Electrical, Resistance, Hermetically Sealed, General Specification for
MS28009	- Indicator – Temperature, Multi-Function, Electrical, Resistance, Single, Clamp Mounted, -70 to +150 Degrees C, 2-Inch Size

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	- Identification Marking of U.S. Military Property

(Copies of these documents are available online at https://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 General. The requirements specified in MIL-I-7070, except qualification requirements, shall be applicable as requirements of this specification. Additional requirements shall be as specified herein.

3.2.1 Precedence. In the event the requirements of MIL-I-7070 and this specification conflict, the requirements of this specification shall govern.

3.2.2 Other performance requirements. The following performance requirements are in addition to those required by MIL-I-7070.

3.2.2.1 Fogging. There shall be no evidence of internal water or oil condensate when subjected to the test of 4.3.2.8. The fogging test shall be part of the sampling plan A testing.

3.3 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.4 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.5 Construction.

3.5.1 Case. The indicator case shall conform to MS28009.

3.5.2 Dial. The dial shall conform to Figure 1. The diameter of the dial as measured across the outer ends of the graduations shall not be less than 1.75 inches.

3.5.2.1 Scale arc. The indicator shall provide a minimum scale arc angle of 250 degrees. The mechanism shall not be geared to obtain the specified scale arc.

3.5.3 Pointer. The pointer shall be attached to the indicator movement. The pointer shall not touch the dial or cover glass. The shape of the pointer shall facilitate reading of any point in the range of the dial. Not less than the outer half of the visible portion of the pointer shall be flat or lusterless white conforming to Color 37875 of FED-STD-595 or fluorescent luminescent as applicable.

3.5.3.1 Pointer overswing. The pointer overswing shall not be less than 3 angular degrees beyond each end of the scale.

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3.6 Markings.

3.6.1 Dial marking. The markings shall be durable and in accordance with Figure 1. The dimensions and colors of the dial markings shall be as listed in Table 1. The flat or lusterless white shall be Color 37875 of FED-STD-595. The flat or lusterless black shall be Color 37038 of FED-STD-595.

TABLE I – Dial Markings.

Marking	Height or Length Inch ±0.005	Width of Line or Graduation Inch ±0.005	Finish
Numerals	0.125	0.020	Fluorescent Luminescent or Flat or Lusterless White
50°C Graduations	0.156	0.020	
10°C Graduations	0.094	0.015	
5°C Graduations	0.063	0.010	
Lettering "°C"	0.125	0.020	
Mask Lettering	0.125	0.020	
Decal Lettering	0.125	0.020	
Visible Portion of Pointer	---	---	
MS Part Number	0.062	0.010	Flat or Lusterless Black
Background of Dial and Mask	---	---	

3.6.2 Mask. The mask shall be either a disk of suitable metal, or a decal affixed firmly and permanently to the center of the inside of the cover glass and shall contain the word "OIL."

3.7 Weight. The weight of the complete indicator shall not exceed 1 pound.

3.8 Electrical receptacle. An electrical receptacle shall be provided and located as shown on MS28009. The receptacle shall be wired as follows:

Terminal	Connection
A	Positive battery (24 VDC)
B	Resistance bulb
C	Common – negative battery (24 VDC) and resistance bulb
D	Open

3.9 Identification of products. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.10 Instrument identification decals. The indicators "MS28009-2" and "MS28009-4" shall be provided with identification decals as shown in Figure 1. The decal shall be larger in diameter than the mask, but not of a size to interfere with readability of the indicator. The decal

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cover glass of the indicator shall be of the water-applied type and shall be suitable for application to the cover glass of the indicator. The decal, when mounted on the cover glass of the indicator, shall be capable of meeting the tests specified in this specification, without peeling, blistering, fading, loss of adhesion, checking, softening, or hardening.

3.10.1 Decal packaging. Each decal set shall be packaged in an envelope marked: "THIS ENVELOPE CONTAINS INDENTIFICATION DECALS". Complete and detailed instructions for mounting the decals on the cover glass of the indicator shall be furnished with each decal set.

4. VERIFICATION

4.1 General specification. The indicator shall be subjected to the verification provisions in accordance with MIL-I-7070 except as specified herein.

4.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection – in accordance with MIL-I-7077 qualification testing
b. Conformance inspection – in accordance with MIL-I-7077 quality conformance testing

4.2 Test conditions.

4.2.1 Test voltage. The test voltage shall be 28.5 ± 0.5 VDC, unless otherwise specified.

4.2.2 Test tolerance. The tolerances specified in Table II, not enclosed in parentheses, are applicable only when the indicators are subjected to all the tests in the sequence specified herein. The tolerances specified in Table II, enclosed in parentheses, are applicable only when the indicators are subjected to a specific test independent of sequence.

4.3 Test methods. Unless otherwise specified, the test listings below are only the tolerances or requirements of the tests specified in MIL-I-7070. Lack of listing herein does not waive the test requirement.

4.3.1 Individual tests. Testing shall be in accordance with MIL-I-7070 individual tests.

4.3.1.1 Scale error at room temperature. The test resistance specified in Table II shall be applied to the resistance bulb terminals of the indicator. The test voltage shall be applied and the scale errors at the test points specified in Table II shall not exceed the tolerance specified in Table II.

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TABLE II – Test Tolerances.

Test Point	Test Resistance	Scale Error at Room Temp	Reduced Voltage	Scale Error at Low Temp	Scale Error at High Temp	Scale Error at Room Temp After Vibration
°C	Ohms	Tol ± °C	Tol ± °C	Tol ± °C	Tol ± °C	Tol ± °C
-70	68.27	4.4 (5.0)	3.0 (3.0)	6.6 (6.6)	6.6 (6.6)	6.6 (6.6)
-30	80.56	4.4 (4.4)	3.0 (3.0)	6.6 (6.6)	6.6 (6.6)	6.6 (6.6)
0	90.38	4.4 (4.4)	2.0 (2.0)	6.6 (6.6)	6.6 (6.6)	6.6 (6.6)
+30	100.91	4.4 (4.4)	2.0 (2.0)	6.6 (6.6)	6.6 (6.6)	6.6 (6.6)
80	120.36	4.4 (4.4)	2.0 (2.0)	6.6 (6.6)	6.6 (6.6)	6.6 (6.6)
120	137.78	4.4 (4.4)	3.0 (3.0)	6.6 (6.6)	6.6 (6.6)	6.6 (6.6)
150	151.91	4.4 (5.0)	3.0 (3.0)	6.6 (6.6)	6.6 (6.6)	6.6 (6.6)

Test Point	Test Resistance	Scale Error at Room Temp After Open and Short Circuit	Scale Error With Bulb Lead Resistance	Extreme Temperature Exposure		
				Scale Error at Low Temp	Scale Error at Room Temp After Low Temp	Scale Error at Room Temp After High Temp
°C	Ohms	Tol ± °C	Tol ± °C	Tol ± °C	Tol ± °C	Tol ± °C
-70	68.27	6.6 (6.6)	3.0 (3.0)	9.5 (6.6)	6.6 (4.4)	6.6 (4.4)
-30	80.56	6.6 (6.6)	3.0 (3.0)	9.5 (6.6)	6.6 (4.4)	6.6 (4.4)
0	90.38	6.6 (6.6)	3.0 (3.0)	7.5 (6.6)	6.6 (4.4)	6.6 (4.4)
+30	100.91	6.6 (6.6)	3.0 (3.0)	7.5 (6.6)	6.6 (4.4)	6.6 (4.4)
80	120.36	6.6 (6.6)	3.0 (3.0)	7.5 (6.6)	6.6 (4.4)	6.6 (4.4)
120	137.78	6.6 (6.6)	3.0 (3.0)	9.5 (6.6)	6.6 (4.4)	6.6 (4.4)
150	151.91	6.6 (6.6)	3.0 (3.0)	9.5 (6.6)	6.6 (4.4)	6.6 (4.4)

4.3.1.2 Position error. The position errors of the indicator shall not exceed 3°C.

4.3.1.3 Friction error. The friction errors of the indicator shall not exceed 3°C.

4.3.1.4 Sealing. When a mass spectrometer, or an equivalent quantitative measuring device of at least equal sensitivity is used, the total leak rate of the indicator shall not exceed one micron cubic foot per hour.

4.3.1.5 Insulation resistance. The insulation resistance shall be a minimum of 20 mega-ohms.

4.3.2 Sampling plan A tests. Testing shall be in accordance with MIL-I-7070 sampling plan A tests plus the fogging test of 4.3.2.8.

4.3.2.1 Reduced voltage operation. The indicator shall be subjected to a scale error at room temperature with an applied voltage of 22.5 volts at the test points specified in Table II.

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The indicators at the reduced voltage shall not differ from the indicators with 28.5 volts applied by more than the tolerances specified in Table II.

4.3.2.2 Scale error at low temperature. The scale error shall not exceed the tolerances specified in Table II.

4.3.2.3 Scale error at high temperature. The scale error shall not exceed the tolerances specified in Table II.

4.3.2.4 Magnetic effect. The magnetic compass shall not deflect more than 5 degrees.

4.3.2.5 Resistance bulb current. A total resistance of 100 ohms shall be applied across the resistance bulb terminals of the indicator, the test voltage applied and the resistance bulb circuit current measured. The current shall not exceed 22 milliamperes.

4.3.2.6 Vibration. The pointer oscillation shall not exceed 2°C. The pointer variation shall not exceed 2°C. The scale errors after vibration shall not exceed the tolerances specified in Table II.

4.3.2.7 Open and short circuit. The scale errors after open and short circuit shall not exceed the tolerances specified in Table II.

4.3.2.8 Fogging. The indicator shall be placed in a 71 ± 2°C controlled ambient temperature for a minimum of 1 hour. At the end of this period and while still at this temperature, an ice cube shall be rubbed on the indicator glass face for 1 minute ± 5 seconds. The glass shall be dried (do not use compressed air). The indicator shall meet the condensate requirement of 3.2.2.1.

4.3.3 Sampling plan B tests. Testing shall be in accordance with MIL-I-7070 sampling plan B tests.

4.3.3.1 Bulb lead resistance effect. The indicator shall be connected as shown on Figure 2 and shall be subjected to a scale error at room temperature test. The indicator with bulb lead resistance shall not differ from the indicator without bulb lead resistance by more than the tolerances specified in table II.

4.3.3.2 Response time. When the indicator is subjected to the response time test, the response time shall not exceed 3 seconds.

4.3.3.3 Damping factor. When the indicator is subjected to the damping factor test, the damping factor shall not be less than 1.6.

4.3.3.4 Power consumption. When the indicator is subjected to the power consumption test, the power consumption shall not exceed 2 watts.

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4.3.3.5 Extreme low temperature exposure. The scale errors at low temperature shall not exceed the tolerances specified in Table II. The scale errors at room temperature after low temperature exposure shall not exceed the tolerances specified in Table II.

4.3.3.6 High temperature exposure. The scale errors at room temperature after high temperature exposure shall not exceed the tolerances specified in Table II.

4.3.3.7 Thermal shock and fogging. No damage to the indicator shall result from the thermal shock test.

4.3.3.8 Salt fog. The indicator shall be subjected to and meet the requirements of the sealing test, scale error and insulation leakage tests following the salt fog exposure.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the inventory control point's packaging activities within the military service or defense agency or within the military service's system command. Packaging data retrieval is available from the managing military department's or defense agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The electrical resistance temperature indicator covered by this specification is intended to indicate temperature in the range of -70°C to +150°C, such as oil, coolant, surface and free air.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Designation (PIN) of indicator required (see 1.3).
c. Ordering data in accordance with MIL-I-7070.
d. If FAT required (see 3.3).
e. Packaging requirements (see 5.1).

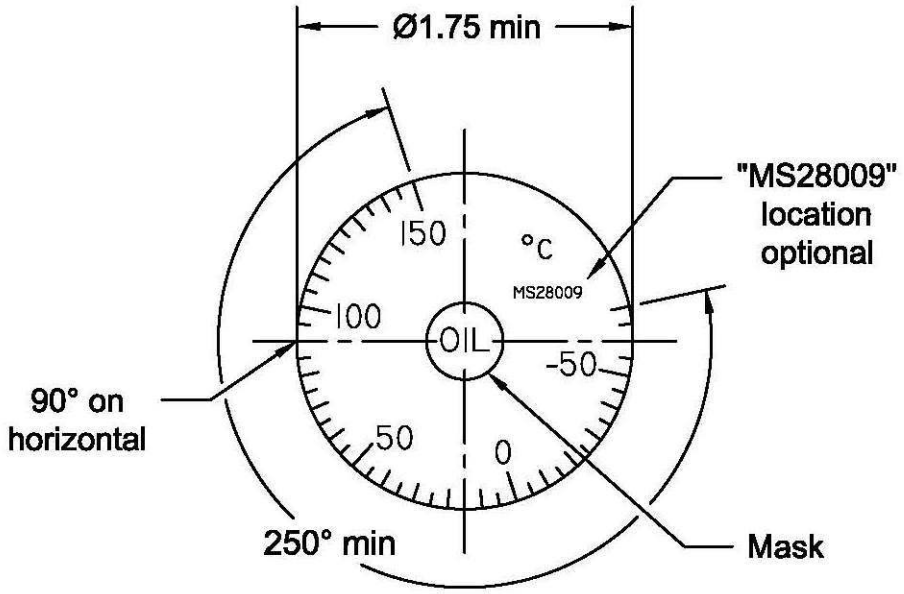
6.3 Bezel. The temperature indicator covered by this specification can be used on instrument panels requiring MS33638 case mounting by the application of an instrument mounting bezel in accordance with MS28055.

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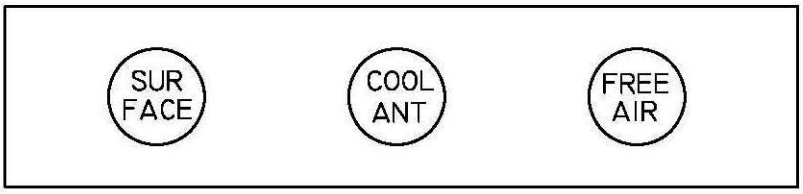
6.4 Subject term (key word) listing.

air
aircraft
cabin

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



Dial face with mask



Mask decalcomania

FIGURE 1. Dial face and mask decalcomania details.

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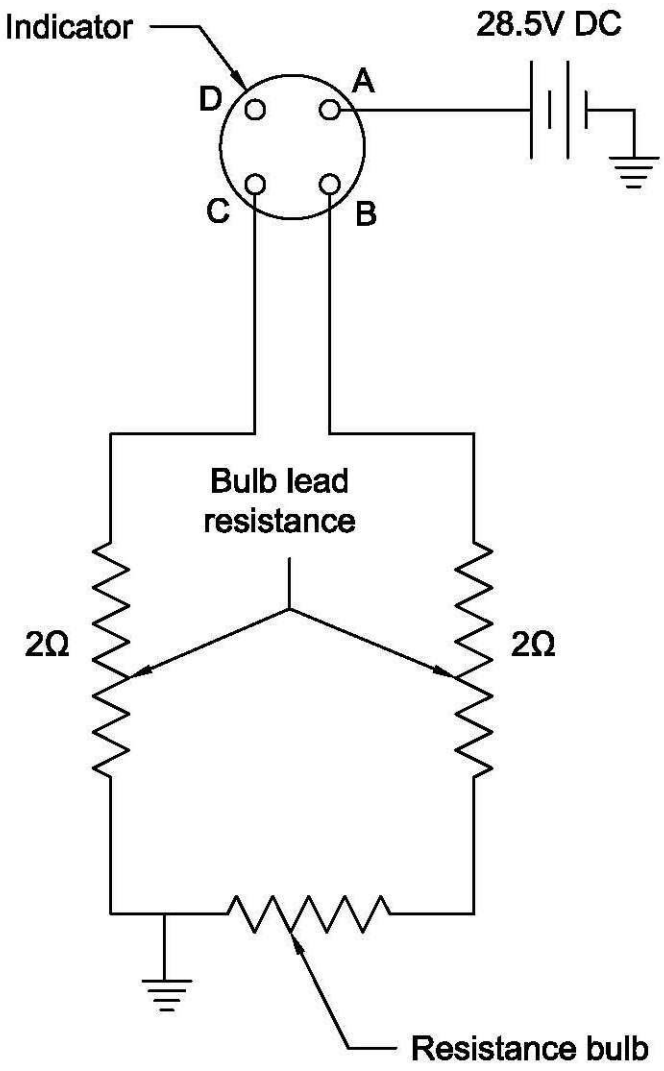


FIGURE 2. Bulb lead resistance test wiring diagram

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